P-P-553B
October 20, 1977
SUPERSEDING
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August 5, 1963

FEDERAL SPECIFICATION

POLISH, FURNITURE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers liquid polishes for cleaning and polishing wood and metal furniture (see 6.1).

1.2 Classification.

1.2.1 Types and classes.

Type I - Liquid emulsion cleaner and polish with oil.
Type II - Liquid emulsion cleaner and polish.

Class 1 - Wax emulsion.
Class 2 - Wax emulsion with silicone.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage And Lighter.
PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller For Other Than Medicinal Products; Packaging and Packing Of.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Hazardous Substances and Articles; Administration and Enforcement Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 56 - Flash Point by Tag Closed Tester, Test for.
D 86 - Distillation of Petroleum Products, Test for.
D 94 - Saponification Number of Petroleum Products, Test for.
D 2834 - Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer-Emulsion Floor Polishes, Test for.
E 70 - pH of Aqueous Solutions with the Glass Electrode, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The products shall have no objectionable or offensive odors.

3.1.1 Type I. The liquid emulsion cleaner and polish with oil shall consist of a stable emulsion of oil in water and shall be free from abrasives and dyes.

3.1.2 Type II, class 1. The liquid emulsion cleaner and polish with wax shall be a stable system of wax and volatile aliphatic hydrocarbons, suitably emulsified in water. The emulsion shall be a free-flowing liquid which can be easily applied and spread with a cotton cloth. The cleaner-polish shall impart a glossy lustre to wood or metal furniture when applied as directed.

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3.1.3 Type II, class 2. The liquid polish shall be a stable system of wax, silicone oil and volatile aliphatic hydrocarbons, suitably emulsified in water. The emulsion shall be a free-flowing liquid which can be easily applied and spread with a cotton cloth. The cleaner-polish shall remove dirt and tarnish and shall impart a glossy lustre to wood or metal furniture when applied as directed.

3.2 Chemical and physical requirements. The polishes shall conform to the requirements in table I when tested as specified in table II.

TABLE I. Chemical and physical requirements

Property	Type I		Type II - Class 1		Type II - Class 2	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Oil content, percent.....	40.0	---	---	---	---	---
Oil and water content, percent..	98.0	---	---	---	---	---
Free caustic alkali.....	---	None	---	None	---	None
Saponification number.....	---	15	---	---	---	---
Nonvolatile matter, percent.....	---	---	4.0	8.0	6.0	12.0
Silicone oil, percent.....	---	---	---	---	2.0	4.0
Volatile hydrocarbon, percent...	---	---	20.0	40.0	20.0	40.0
Flash point, °F.....	---	---	95	---	95	---
Distillation range °C.....	---	---	140	221	140	221
pH value.....	---	---	6.0	9.5	6.0	9.5
Abrasiveness.....	---	None	---	None	---	None

3.3 Stability properties. The polishes shall show no evidence of settling or break-down of the emulsions when tested as specified in 4.4.14.

3.4 Application. The polishes shall flow freely and spread easily on the surface to which it is applied when tested as specified in 4.4.15.

3.5 Gloss increase.

3.5.1 Type I. The gloss of the panel after polishing shall increase not less than 25 percent when tested as specified in 4.4.16.

3.5.2 Type II, class 1. The gloss of the polished panel shall increase not less than 25 percent when tested as specified in 4.4.16.

3.5.3 Type II, class 2. The gloss of the polished panel shall increase not less than 40 percent when tested as specified in 4.4.16.

3.6 Gloss retention. The gloss of the polished panel shall decrease not more than 5 percent when tested as specified in 4.4.16.

3.7 Labeling. The product labels shall include the directions for application and buffing and a list of the types of surfaces on which the product can be used. The labels shall be in accordance with the Federal Hazardous Substances Act and Regulations (16 CFR 1500).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one classification (see 1.2) and one package size (see 5.1).

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4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Sampling procedure. The sample unit shall be defined as one shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level S-2 and the AQL shall be 4.0 percent defective.

4.3.1.2 Examination procedure. An examination shall be made to determine the compliance with the packing and marking requirements of section 5. The sample units shall be examined for the defects described below and scored accordingly.

Examination of preparation for delivery

Examine	Defects
Packing	Not as specified; odd size or shape of container; insufficient packages inside.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of packing containers or of component materials.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.

4.3.2 Examination of the filled unit container.

4.3.2.1 Sampling procedure. The sample unit shall be defined as one filled container. The sample units shall be selected in accordance with inspection level S-2 and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. An examination shall be made to determine the compliance of the contents and packaging with the requirements of 3.1, 3.7, and section 5. The sample units shall be examined for the defects described below and scored accordingly.

Examination of the filled unit container

Examine	Defects
Contents	Not free flowing; presence of obvious artificial coloring; evidence of separation; rancid or objectionable odor.
Fill	Insufficient.
Packaging	Not as specified (see 5.1).
Marking	Not as specified (see 3.7 and 5.3).

4.4 Testing.

4.4.1 Sampling plan. The sample unit shall be defined as one unit container. Testing shall be performed on a one quart composite made by mixing equal portions of sample units taken from the lot in accordance with inspection level S-1. The lot shall be unacceptable if the composite sample fails to pass any of the tests described in 4.4.2.

4.4.2 Testing procedure. The composite sample shall be tested in accordance with the instructions in table II.

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TABLE II. Instructions for testing the composite sample

Characteristic	Applicable to type	Specification reference		Results reported as	
		Requirement	Test method	Pass or Fail	Numerically to nearest
Oil	I	Table I	4.4.3	--	0.1 percent
Water and oil content	I	Table I	4.4.4	--	0.1 percent
Free caustic alkali	I, II	Table I	4.4.5	X	---
Saponification number	I	Table I	4.4.6	--	1.0 number
Nonvolatile matter	II	Table I	4.4.7	--	0.1 percent
Silicone oil content	II class 2	Table I	4.4.8	--	0.1 percent
Volatile hydrocarbon content	II	Table I	4.4.9	--	0.1 percent
Flash point	II	Table I	4.4.10	--	1.0°F.
Distillation range	II	Table I	4.4.11	--	1.0°C.
pH value	I, II	Table I	4.4.12	--	0.1 pH
Abrasiveness	I, II	Table I	4.4.13	X	---
Stability properties	I, II	3.3	4.4.14	X	---
Application	I, II	3.4	4.4.15	X	---
Gloss increase	I, II	3.5	4.4.16	--	1.0 percent
Gloss retention	I, II	3.6	4.4.16	--	1.0 percent

4.4.3 Oil. Place approximately 100 grams of the sample, accurately weighed, in a beaker and dilute to about 200 ml with distilled water. Heat mixture and acidify with 0.5N hydrochloric acid, using methyl orange as indicator. After oil has collected at the top, carefully add 50 to 100 ml of benzene, transfer to a separatory funnel and wash the benzene solution of the oil with water until free of acids. Draw off the water layer and transfer the benzene layer to a weighed beaker. Evaporate the benzene on the steam bath and dry the oil in an oven at 100°-105°C for 24 hours. Cool in a desiccator and weigh. Calculate the percentage of oil in the sample.

4.4.4 Water and oil content. Distill 100 grams of the sample using a distillation flask connected to a condenser and having a 100 ml graduated cylinder for a fraction receiver. In the graduated receiver collect the distillation fraction produced at 100° ± 1°C (at standard pressure). Calculate the percentage of water using the following formula:

$$\text{Water, percent} = \frac{V \times 0.998}{S} \times 100$$

Where:

V = volume of water in milliliters at 20°C.
S = weight of sample in grams.

4.4.5 Free caustic alkali. A 10 ml sample of the polish shall be digested in 100 ml of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator and, if no pink color results at this point, no free caustic alkali is present. Should a pink color appear, dilute about 10 ml of an original sample of polish with 10 ml of distilled water, boil and test the vapors for ammonia. If present, continue boiling until the ammonia is driven off, cool, add 100 ml of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator. No pink coloration shall result.

4.4.6 Saponification number. The saponification number shall be determined in accordance with ASTM Method D 94.

4.4.7 Nonvolatile matter. The percentage nonvolatile matter (total solids) shall be determined in accordance with ASTM Method D 2834.

4.4.8 Silicone oil (dimethyl siloxane polymer). Accurately weigh 3 to 4 grams of the polish and extract (into the organic layer) three times with 20 ml portions of chloroform using a small separatory funnel. Place the combined chloroform layers into an evaporating dish and dry in a forced air oven maintained at less than 100°C until the volume has been concentrated to about 5 ml. Transfer to a tared platinum or vycor crucible, washing the evaporating dish into the crucible with 10 ml of chloroform. Place in a forced air oven maintained at 105° ± 2.5°C. for 2 hours to remove volatile matter and then cool in a

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desiccator. To reduce subsequent frothing of sample, add 4 or 5 drops of chlorobenzene and agitate or warm contents to effect solution. Add 2 ml of fuming sulfuric acid to the solution at room temperature. Add 0.5 ml of fuming nitric acid immediately following the addition of the sulfuric acid. Heat crucible and contents over a low flame for several hours to accomplish digestion. When the mass has solidified and danger of frothing is past, heat the crucible over a Meker burner to complete expulsion of sulfur trioxide fumes. Place residue in a muffle furnace for 1 hour at 800°C. Cool in a desiccator and weigh residue.

4.4.8.1 Calculation. Calculate percent silicone oil as follows:

$$\text{Percent silicon dioxide} = \frac{\text{weight of residue}}{\text{weight of sample}} \times 100$$

$$\text{Percent dimethyl siloxane polymer (silicone oils)} = \text{percent silicon dioxide} \times \frac{74}{60}$$

4.4.9 Volatile hydrocarbon content. To 500 grams of the furniture polish in a one liter round bottom flask, add sufficient dilute hydrochloric acid to break the emulsion. Add boiling chips to prevent bumping and connect the flask to a water cooled condenser whose outlet is joined to a 500 ml flask by a 105? vented connector tube. Ground glass joints are desirable. Heat carefully to distill over the volatile material using a suitable hot plate, electric mantle, or oil bath. Separate the petroleum hydrocarbons in the distillate from the water with a separatory funnel and transfer it to a tared beaker. Determine the weight of the petroleum hydrocarbon and compute the percentage. Save the separated hydrocarbon for the determinations in 4.4.10 and 4.4.11.

4.4.10 Flash point. The flash point shall be determined in accordance with ASTM Method D 56.

4.4.11 Distillation range. The distillation range shall be determined in accordance with ASTM Method D 86.

4.4.12 pH. The pH shall be determined in accordance with ASTM Method E 70, using the glass electrode.

4.4.13 Abrasiveness. Spread a thin film of polish over a microscope slide. Place another microscope slide over the film of polish. Press the two slides together using a slight pressure and at the same time sliding one glass slide over the other. The movement between the glass slides shall be smooth without any indication of the presence of abrasive.

4.4.14 Stability properties. A graduated glass cylinder about 1 inch in diameter shall be filled with a sample of the polish to a height of 10 inches. The cylinder shall be glass-stoppered and left undisturbed at a temperature of 20° to 25°C for 48 hours. The polish shall remain as a uniform emulsion without settling (layer formation) and without breakdown (appearance of oil) of the emulsion.

4.4.15 Application. Carry out test under condition of 20° to 30°C. Pour a small amount of polish onto a soft cloth. The polish shall flow freely from the container. Apply the polish with the cloth to a finished wood and metal surface. The polish shall spread easily to a thin film. Rub the film of polish with a dry cloth, using no more than moderate pressure. The polished surface shall be free from oiliness or tack and shall have a good lustre.

4.4.16 Gloss increase and retention. Select a clean rubbed-varnished or rubbed-lacquered panel approximately 1 by 2 feet, having a gloss meter reading between 20 and 30, using a Gardner 60° Glossmeter. A minimum of ten readings shall be taken to determine an average, moving and rotating the glossmeter 90° after each reading so that the entire surface shall be represented by the average. This average is the gloss reading of the panel. Apply the polish as specified in 4.4.15 and, directly after buffing, determine the gloss reading. The percentage increase in gloss shall be not less than 25 percent for type I and type II, class 1 and not less than 40 percent for type II, class 2. Store the panel for 24 hours at 25° ± 2°C and 50 percent ± 2 percent relative humidity and again obtain the average gloss reading to determine gloss retention. The gloss of the polish shall not decrease more than 5 percent.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One quart. One quart of one type and class product only shall be furnished in a metal can conforming to PPP-C-96, type V, class 4, exterior coating plan A, or a glass bottle with a screw cap closure normally used for the item or a plastic bottle with a screw cap closure normally used for the item.

5.1.1.2 One gallon. One gallon of one type and class product only shall be furnished in a metal can conforming to PPP-C-96, type V, class 4, exterior coating plan A or a plastic bottle with screw cap closure normally used for the item.

5.1.2 Commercial. Furniture polish shall be packaged in accordance with normal commercial practice in one quart or one gallon containers, as specified (see 6.2).

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Metal cans. Twelve 1-quart or six 1-gallon filled cans shall be packed in accordance with PPP-C-96 for level A packing.

5.2.1.2 Plastic bottles. Twelve 1-quart or six 1-gallon filled plastic bottles shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be fitted with partitions or dividers to separate the bottles and increase the stacking strength of the box. The partitions or dividers shall be 3/8 inch ± 1/8 inch higher than the bottles. The box shall be closed, waterproofed, and banded in accordance with the appendix to PPP-B-636.

5.2.1.3 Glass bottles. Twelve 1-quart filled glass bottles shall be packed in accordance with PPP-G-460 for level A packing.

5.2.2 Commercial. Furniture polish shall be packed in containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification Rules or Uniform Freight Classification Rules. The number of packages packed in a shipping container shall be as specified in the procurement document (see 6.2).

5.3 Marking.

5.3.1 Civil agencies. The unit container shall be marked in accordance with 3.7, the Federal Hazardous Substances Act, and Federal Standard No. 123. The shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. The unit containers shall be marked in accordance with 3.7 and the Federal Hazardous Substances Act. The unit and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The furniture polishes covered by this specification are intended primarily for the cleaning and polishing of furniture and other finished wood and metal surfaces. The polishes are intended for varnished, lacquered, or enameled furniture finishes. They are not intended for exterior or floor surfaces.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Whether the packing and packaging is required to be in accordance with Level A or Commercial requirements (see 5.1 and 5.2).
(d) Size of packaging required (see 5.1).
(e) The number of packages required per shipping container (see 5.2.2).

MILITARY INTEREST:

The Department of Defense has waived coordination interest in this document.

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA
CPSC

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60cents each.

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